Youssef Shalabya

Data Analyst | Data Scientist | Machine Learning Engineer

Location: Benha, Qalyubia — Cairo, Egypt **Phone**: +20 114 224 4215
Email: youssef.shalabya@gmail.com **LinkedIn**: linkedin.com/in/youssef-shalabya **GitHub**: YoussefShalabya

Profile Summary

Motivated Communications and Computer Engineering student with strong hands-on experience in data analytics, machine learning, and artificial intelligence. Proficient in Python, SQL, Power BI, Excel, and Tableau, with the ability to clean, analyze, and visualize complex real-world datasets to extract actionable insights. Seeking opportunities in Data Analysis, Data Science, or Machine Learning roles.

Education

Bachelor of Communications and Computer Engineering

Benha Faculty of Engineering, Benha University, Egypt **2021 – 2026 (Expected)**

- Cumulative Grade (up to last level): Excellent – 87%
- Core studies: Mathematics, Probability & Statistics, Embedded Systems, Signals & Systems, Data Analysis, AI & Machine Learning

Training & Technical Experience

Machine Learning Diploma

Creativa Innovation Hub, Benha **Apr 2026 – May 2026**

Core Machine Learning:

- Fundamentals of supervised and unsupervised learning with Python and Scikit-learn
- Algorithms covered: Linear & Logistic Regression, Decision Trees, Random Forest, SVM, K-Means Clustering, K-Nearest Neighbours
- Hands-on feature engineering, model evaluation, cross-validation, and hyperparameter tuning

Data Analysis Using AI Training

Orange Digital Center × Digital Hub, Dokki, Cairo **Feb 2026 – Apr 2026**

- Leveraged modern AI tools to automate and accelerate data analysis workflows
- Applied AI-assisted techniques for data cleaning, outlier detection, and feature extraction
- Conducted advanced EDA using AI-driven analytics platforms
- Built interactive dashboards integrating AI-generated insights for business decision-making
- Designed predictive analysis models and automated reporting pipelines
- Generated AI-driven insights from diverse datasets across multiple industries

Machine Learning Summer Training

NTI (National Telecommunication Institute) **Jul 2025 – Aug 2025**

- Applied core ML algorithms (regression, classification, clustering) to real-world analytical problems
- Used Python and Scikit-learn to preprocess data, build models, and evaluate performance metrics
- Focused on data-driven decision making using trained predictive models

Artificial Intelligence Track

NTI – Huawei Egyptian Talents Academy (ETA) **Jun 2025 – Jul 2025**

- Python programming for AI applications: data structures, OOP, and libraries (NumPy, Pandas, Matplotlib)
- Mathematics for ML: linear algebra, calculus, probability, and statistics
- Machine learning foundations: supervised, unsupervised, and reinforcement learning concepts
- Deep learning fundamentals: neural networks, activation functions, forward and backpropagation

Data Analytics Trainee

Digital Egypt Pioneers Initiative (DEPI) – Ministry of Communications **Oct 2024 – May 2025**

- Performed comprehensive data cleaning and preprocessing on large, messy real-world datasets using Python (Pandas) and Excel
- Handled missing values, duplicates, outliers, and inconsistent data formats to ensure data quality
- Conducted thorough EDA to uncover patterns, trends, and correlations within datasets
- Applied statistical methods: descriptive statistics, hypothesis testing, and correlation analysis
- Built dynamic dashboards and reports using Excel, Tableau, and Python (Matplotlib, Seaborn)

Key Skills and Abilities

Data Analysis & Business Intelligence:

- Power BI (DAX, Power Query, interactive dashboards, KPI tracking)
- Microsoft Excel (Pivot Tables, advanced formulas, dynamic dashboards, data validation)
- Tableau (data visualization, calculated fields, story dashboards)
- SQL (complex queries, joins, aggregations, subqueries, window functions)
- Data Cleaning, Feature Engineering, Exploratory Data Analysis (EDA)

Programming & Libraries:

- Python: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- C / C++ (embedded systems, algorithmic problem solving)
- HTML/CSS (basic web development)

Machine Learning & AI:

- Supervised Learning: Regression, Classification, Decision Trees, Random Forest, SVM
- Unsupervised Learning: K-Means Clustering, Dimensionality Reduction
- Deep Learning: Neural Networks fundamentals, Backpropagation
- Model evaluation: Cross-validation, Confusion Matrix, ROC-AUC, RMSE

Soft Skills:

- Problem Solving, Critical Thinking, Analytical Mindset
- Communication & Presentation Skills, Teamwork & Collaboration
- Time Management, Adaptability, Attention to Detail

Projects

DEPI Data Analytics Dashboards

Python · SQL · Excel · Power BI **2025**

- Built end-to-end analytics dashboards from real-world datasets during the DEPI national program
- Performed data cleaning, EDA, and statistical analysis to extract actionable business insights
- Delivered polished visualizations for case study presentations to program stakeholders

Power BI Portfolio Dashboard

Power BI · DAX · Power Query **In Progress**

- Developing a series of interactive dashboards covering sales analysis, KPI tracking, and operational reporting
- Implementing clean data models with optimized DAX measures and Power Query transformations

ML Models — Creativa Hub

Python · Scikit-learn · Pandas · Streamlit **In Progress**

- Building classification, regression, and clustering models on real datasets as part of the ML Diploma
- Full pipeline from raw data preprocessing to trained model deployment via Streamlit web app

Excel Analytics Projects

Excel · Pivot Tables · Charts **In Progress**

- Creating business reporting dashboards using advanced Excel features for data-driven decision making

Certifications & Achievements

- Google Data Analytics Professional Certificate — DEPI / Ministry of Communications (2025)
- Machine Learning Diploma — Creativa Innovation Hub, Benha (2026)
- Data Analysis Using AI — Orange Digital Center × Digital Hub (2026)
- Artificial Intelligence Track — NTI × Huawei Egyptian Talents Academy (2025)
- Machine Learning Summer Training — NTI (2025)
- Linux Administration Bootcamp — Linux-Plus Information Systems (2025)
- SQL for Data Analysis — DataCamp (2025)
- Intro to Data Analysis — DataCamp (2025)
- Excel Dashboard Bootcamp — Creativa Innovation Hub (2024)

Volunteering Experience

Volunteer

IEEE BUB Student Branch **Sep 2023 – Oct 2024**

- Assisted in organizing workshops and technical seminars for engineering students
- Supported communication between speakers and attendees, ensuring smooth event flow

Languages

Arabic: Native

English: Very Good